

SEMENIC, Romania

WMO: 153150

Lat: 45.1814N

Lon: 22.0558E

Elev: 1433

StdP: 85.25

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-13.0	-22.2	0.6	-7.8	-18.1	0.9	-6.5	15.7	-7.1	14.2	-5.2	5.9	130	0.729

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	22.6	15.7	21.1	15.2	19.6	14.6	16.9	20.5	16.2	19.5	15.4	18.7	4.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	13.4	18.2	14.9	12.7	17.5	14.1	12.0	16.9	53.3	20.5	50.9	19.8	48.6	18.8	20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.7	12.5	10.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-4.9	-3.2	-1.4	3.6	8.8	12.8	14.2	14.6	10.1	5.5	1.5	-2.7
	DBStd	8.09	4.92	4.31	4.39	5.50	3.66	3.55	3.17	3.52	4.35	4.83	4.75	4.61
	HDD10.0	2330	461	370	353	199	66	14	4	5	53	154	257	394
	HDD18.3	4897	719	603	612	441	296	166	133	123	247	398	506	653
	CDD10.0	489	0	0	0	8	28	99	133	147	57	14	1	0
	CDD18.3	15	0	0	0	0	0	2	4	7	2	0	0	0
	CDH23.3	24	0	0	0	0	0	1	7	13	3	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	4.7	4.9	5.8	4.9	4.9	4.7	3.8	3.6	3.9	4.5	4.6	5.7	5.2
Precipitation	PrecAvg	1175	66	65	60	90	139	181	146	124	92	67	68	75
	PrecMax	1683	175	172	193	218	304	552	310	335	256	258	161	148
	PrecMin	672	8	3	8	22	26	72	36	20	6	7	0	3
	PrecStd	284	43	41	36	44	69	100	72	83	60	63	37	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	8.7	10.7	17.0	19.9	22.2	24.1	25.0	22.8	18.1	13.8	9.5
		MCWB	1.4	4.8	6.1	9.3	13.2	16.1	16.1	16.8	16.0	10.4	4.9	3.3
	2%	DB	4.4	5.4	7.8	15.2	17.2	20.8	22.1	22.9	19.5	16.0	10.8	6.6
		MCWB	1.1	3.0	4.0	9.2	11.7	16.1	15.8	15.5	13.7	10.3	4.3	2.4
	5%	DB	3.1	3.7	6.2	12.9	15.5	19.5	20.5	21.2	17.5	14.0	8.9	4.7
		MCWB	1.2	1.7	2.9	7.7	10.8	15.2	14.8	15.2	13.0	8.6	4.9	2.0
	10%	DB	1.5	2.3	4.4	11.0	14.0	18.1	18.9	19.5	16.0	12.2	7.2	3.0
		MCWB	0.2	0.9	1.7	6.9	10.1	14.3	14.5	14.5	12.0	8.3	5.2	0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	5.9	6.6	11.5	13.8	18.2	17.5	18.5	15.9	12.6	8.0	5.5
		MCDB	4.8	7.7	11.1	14.7	17.5	20.3	21.2	22.0	21.4	15.7	9.4	7.5
	2%	WB	2.8	3.9	4.8	9.6	12.6	16.8	16.6	16.8	14.6	11.3	7.2	3.4
		MCDB	3.7	4.9	7.1	13.5	16.0	19.7	20.1	20.7	18.9	14.5	8.2	5.2
	5%	WB	1.4	2.2	3.5	8.5	11.5	15.9	15.8	16.0	13.3	10.0	6.4	2.4
		MCDB	2.1	3.1	5.5	12.3	14.4	18.7	19.2	19.5	16.5	12.7	7.4	4.4
	10%	WB	0.3	1.2	2.3	7.2	10.5	14.9	15.1	15.2	12.4	8.8	5.6	1.4
		MCDB	1.2	1.8	3.9	10.8	13.3	17.6	18.3	18.6	15.1	11.5	6.7	3.0
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.0	5.3	6.0	5.6	6.0	5.9	5.9	5.2	5.4	4.8	4.7
		MCDBR	6.2	6.3	7.7	8.1	7.4	7.8	7.7	7.1	7.9	7.2	5.8	
		MCWBR	4.9	4.5	4.5	4.8	4.6	5.2	5.1	4.6	4.3	5.2	4.3	3.7
	5% WB	MCDBR	5.5	5.8	6.6	7.4	6.5	7.1	6.8	7.0	6.6	7.0	4.7	5.9
		MCWBR	4.7	4.3	4.5	4.5	4.6	5.4	4.9	4.6	4.3	5.3	3.6	4.8
Clear-Sky Solar Irradiance	taub	0.272	0.294	0.324	0.365	0.357	0.351	0.360	0.370	0.346	0.317	0.294	0.273	
	taud	2.430	2.379	2.385	2.288	2.344	2.380	2.348	2.320	2.380	2.473	2.499	2.470	
	Ebn at Noon	853	894	910	894	911	915	901	878	873	856	820	815	
	Edh at Noon	70	90	103	124	121	118	120	119	103	81	66	61	
All-Sky Solar Radiation	RadAvg	1.27	2.08	3.27	4.49	5.41	6.04	6.02	5.40	3.81	2.60	1.54	1.05	
	RadStd	0.16	0.28	0.44	0.48	0.50	0.56	0.47	0.52	0.40	0.30	0.17	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (67 neighbors)	+0.66	N/A	N/A	N/A	N/A	N/A	-110	-173	+118	+62

Nomenclature: See separate page